Universiti Tun Hussein Onn Malaysia

UNIVERSITI TUN HUSSEIN ONN MALAYSIA

FINAL EXAMINATION
SEMESTER I
SESSION 2024/2025

COURSE NAME	:	DISCRETE STRUCTURE
COURSE CODE	:	BIC 10103
PROGRAMME CODE	:	BIS/BIP/BIW/BIM
EXAMINATION DATE	:	FEBRUARY 2025
DURATION	:	3 HOURS
INSTRUCTIONS	:	1. ANSWER ALL QUESTIONS 2. THIS FINAL EXAMINATION IS CONDUCTED VIA ☐ Open book ☒ Closed book 3. STUDENTS ARE **PROHIBITED** TO CONSULT THEIR OWN MATERIAL OR ANY EXTERNAL RESOURCES DURING THE EXAMINATION CONDUCTED VIA CLOSED BOOK

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THIS QUESTION PAPER CONSISTS OF **FOUR (4)** PAGES

Q1 (a) Express the following English statements using appropriate predicates, quantifiers and logical connectives, with the domain of discourse consisting of all students at your school.

(i) There is a student at your school who can speak Russian but does not know C++.

(2 marks)

(ii) Every student at your school can either speak Russian or knows C++.

(2 marks)

(iii) No student at your school can speak Russion or knows C++.

(2 marks)

(b) Find the Disjunctive Normal Form (DNF) for the following Conjunctive Normal Form (CNF) by constructing a truth table.

$$(p \vee q \vee r) \wedge (\neg p \vee q \vee \neg r)$$

(6 marks)

(c) Construct a truth table for the following proposition.

$$\Big(\big((p \vee q) \wedge (p \rightarrow r)\big) \wedge (q \rightarrow r)\Big) \rightarrow r$$

(7 marks)

(d) Based on answer in **Q1(c)**, determine whether it is tautology, contradiction, or contingency.

(1 mark)

Q2 Prove that $3^n - 1$ is a multiple of 2 for all positive integers n using mathematical induction.

(10 marks)

Q3 (a) Let R be the relation on the set of ordered pairs of positive integers, defined such that $((a, b), (c, d)) \in R$ if and only if $a + d = b + c$. Show that R is an equivalence relation.

(9 marks)

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(b) Let $A = \{1,2,3,4\}$ and functions F, G and H be defined with the domain and codomain of A as follows:

$F(1) = 3, F(2) = 2, F(3) = 2$ and $F(4) = 4$

$G(1) = 1, G(2) = 3, G(3) = 4$ and $G(4) = 2$

$H(1) = 2, H(2) = 4, H(3) = 1$ and $H(4) = 3$

Find the following,

(i) $F \circ G$

(3 marks)

(ii) $H \circ F$

(3 marks)

(iii) $G \circ H$

(3 marks)

Q4 (a) Solve the recurrence relation $a_n = a_{n-1} + 6a_{n-2}$ for $n \geq 2$, where $a_0 = 3$ and $a_1 = 6$.

(10 marks)

(b) A deposit of RM100,000 is made into an investment fund at the beginning of the year. At the end of each year, two dividends are awarded. The first dividend is 20% of the amount in the account for that year. The second dividend is 45% of the amount in the account from the previous year.

(i) Find a recurrence relation for the sequence $\{Pn\}$, where Pn represents the amount in the account at the end of n years, assuming that no money is ever withdrawn.

(3 marks)

(ii) Determine the total amount in the account after n years, if no money has been withdrawn.

(2 marks)

Q5 (a) Suppose there is a group of 5 people (A, B, C, D, and E) with the following pairs acquaintances:

- A and C
- A and D
- B and C
- C and D
- C and E

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(i) Draw a graph G to represent this situation.

(2 marks)

(ii) Show the adjacency matrix for G.

(5 marks)

(b) Five computers are connected in a network, and the provided adjacency matrix shows the cable connections between these computers.

$$\begin{bmatrix} 0 & 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 \end{bmatrix}$$

Based on the given adjacency matrix, analyze whether the network is a Eulerian or Hamiltonian.

(5 marks)

(c) List the Eulerian paths contained in the following graph.

$V = \{1, 2, 3, 4\}$
$E = [(1,2), (1,3), (1,4), (2,3), (3,4)]$

(8 marks)

- END OF QUESTIONS -

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